

Malhar Gudekar

(217) -766-0681 — gudekar2@illinois.edu — LinkedIn — Portfolio — GitHub

EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

• **Analytical:** Data Analysis, Exploratory Data Analysis, Statistical Analysis, Trend Analysis, Root Cause Analysis, KPI Reporting, Data Validation, Data Cleaning, Anomaly Detection, ETL, Predictive Analytics

• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j

• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization

• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

• Owned end-to-end design and development of **cross-platform system integrating** wearable data for 100+ users, enabling scalable **data ingestion** and analytics

• Designed and optimized **PostgreSQL architecture** for high-volume, production-scale datasets, improving query performance by 38%

• Built robust, production-ready **Python** pipeline with retrieval and context summarization, improving multi-turn response accuracy by 43%

• Developed **backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Designed and deployed highly **scalable NLP pipelines in Python** to process high-volume datasets, reducing processing latency by 41%

• Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection

• Developed production-grade **APIs** and deployment workflows to improve system reliability and reproducibility

• Enabled large-scale data processing across social media streams using **distributed system architecture** principles

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

• Built and deployed scalable, microservices-based Retrieval-Augmented Generation (**RAG**) system using **FastAPI**, **PostgreSQL**, and vector embeddings

• Designed efficient, fault-tolerant **ETL** pipelines to process and index large document datasets for low-latency retrieval

• Improved system performance by 64% through optimized indexing strategies and pipeline efficiency improvements

• Developed backend services enabling real-time, high-performance querying of unstructured data

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

• Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by 40% and improving visibility into operational performance across transportation services

• Processed and optimized large-scale operational datasets using **Python** and **PySpark** to identify failure patterns, trends, and performance issues

• Developed SQL queries to map operational dependencies, analyze business processes, and detect data quality issues, accelerating root-cause analysis and identifying efficiency improvements by 26%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

• Developed and maintained 20+ dashboards across Sales, Finance, and Operations using SQL and DAX, documenting processes and surfacing data-driven insights to identify revenue opportunities for enterprise clients

• Automated recurring Excel and SharePoint reporting workflows using VBA, documenting process improvements and implementing validation logic, reducing report preparation time by 42%

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

• Led cross-functional team of 6 to deliver product updates, presenting performance insights and actionable recommendations to stakeholders using data-driven analysis and structured goal-setting

• Tracked sprint milestones and QA issues through Agile JIRA workflows, performing user acceptance testing on new features and maintaining process documentation to ensure on-time delivery

PROJECTS

FMCG Sales Demand Forecasting

March 2025

View Project

• Analyzed FMCG **sales demand data** across categories and locations to identify trends, seasonal patterns, and key drivers impacting inventory

• Built and compared forecasting models including **ARIMA**, **SARIMA**, and regression to evaluate demand patterns and improve forecast accuracy using RMSE and MAPE

• Derived actionable business insights on promotions, pricing, and stock behavior, identifying revenue opportunities and enabling better inventory planning to reduce operational risk